

SAP S/4HANA SECURITY

Chapter 3 – Advanced Fiori Security & S/4HANA Authorization

Q134. What are CDS Views?

Interviewer's Objective

The interviewer wants to know whether you understand how modern S/4HANA applications access data.

Executive Answer

CDS (Core Data Services) Views are virtual data models built on the SAP HANA database that expose business data for applications, reporting, analytics, and APIs.

Unlike traditional SAP reports that directly read database tables, modern S/4HANA applications—including many Fiori apps—retrieve data through CDS Views.

They provide:

- Semantic business models
 - Better performance
 - Reusable data definitions
 - Embedded analytics
 - Security integration
-

Architecture

SAP HANA Database



CDS View



OData Service



Fiori Application

Interview Tip

Many Fiori applications never access tables directly.

They access CDS Views.

Q135. What is CDS Authorization?

Executive Answer

CDS Authorization controls access to records returned by CDS Views.

Instead of only checking whether a user can execute an application,

CDS Authorization determines:

Which business records can be returned.

Example

Production Manager

↓

Plant = 1000

↓

Only production orders belonging to Plant 1000 are returned.

Benefits

- **Row-level security**
 - **Better performance**
 - **Centralized authorization**
 - **Embedded analytics protection**
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Q136. What is DCL (Data Control Language)?

Architect-level interview question.

Executive Answer

DCL (Data Control Language) defines authorization rules for CDS Views.

Think of it as:

SQL controls what data exists.

DCL controls which users may see that data.

Example

```
define role Z_SALES_MANAGER {  
  grant select on ZI_SALES  
    where SalesOrganization = aspect pfcg_auth(...);  
}
```

This ensures users only retrieve data for authorized sales organizations.

Why It Matters

Without DCL, a user might launch a Fiori app successfully but still see data they should not be allowed to access.

Q137. What is Embedded Analytics?

Executive Answer

Embedded Analytics is the native analytical capability built into S/4HANA.

It allows operational reporting directly within transactional applications.

Examples include:

- **Sales dashboards**
- **Procurement KPIs**
- **Inventory analysis**
- **Financial reports**
- **Production performance**

Security for Embedded Analytics relies on:

- CDS Authorization
 - DCL
 - Backend authorization objects
 - Organizational values
-

Q138. How is Security handled for Embedded Analytics?

Executive Answer

Security is layered.

User

↓

Fiori App

↓

OData

↓

CDS View

↓

DCL

↓

Authorization Objects

↓

Database

Each layer contributes to protecting analytical data.

Example

A Plant Manager should only see KPIs for:

Plant

1000

not every manufacturing plant globally.

Q139. Explain RAP Authorization.

One of the newest Architect-level questions.

Executive Answer

RAP (RESTful ABAP Programming Model) is SAP's modern development framework for cloud-ready applications.

Authorization in RAP consists of:

- **Authentication**
- **Authorization Checks**
- **CDS Security**
- **Business Object Authorization**
- **Instance Authorization**

Unlike classic ABAP, RAP integrates security directly into business objects and services.

Interview Tip

If asked about RAP, mention:

- **CDS**
 - **OData**
 - **Business Objects**
 - **Authorization Behavior**
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Q140. How do you secure a Custom Fiori Application?

This question appears frequently in S/4HANA projects.

Executive Answer

I follow a layered security approach.

Step 1

Authentication

IAS

SSO

Entra ID

Step 2

Launchpad Authorization

Catalog

Target Mapping

Spaces

Pages

Step 3

OData Security

Activate required services.

Restrict access.

Step 4

Backend Authorization

Authorization Objects

Organizational Levels

Business Roles

Step 5

CDS Authorization

Implement DCL.

Step 6

Testing

- **SU53**
 - **STAUTHTRACE**
 - **Functional validation**
 - **Penetration testing (if applicable)**
-

Q141. Explain Launchpad Content Manager.

Executive Answer

Launchpad Content Manager is the central administration tool for Fiori Launchpad content.

It is used to manage:

- **Catalogs**
- **Spaces**
- **Pages**
- **Business catalogs**
- **App assignments**
- **Target mappings**

Compared to older Launchpad Designer tools, it provides a more modern and scalable administration experience.

Q142. How do you design Fiori Roles?

Executive Answer

My methodology includes:

Step 1

Business workshop.

↓

Step 2

Identify personas.

↓

Step 3

Identify required applications.

↓

Step 4

Create business catalogs.

↓

Step 5

Assign spaces/pages.

↓

Step 6

Configure backend authorizations.

↓

Step 7

Validate OData.

↓

Step 8

Run SoD Analysis.

↓

Step 9

Testing.

↓

Step 10

Production deployment.

Q143. Explain Authorization Defaults (SU24) for Fiori.

Executive Answer

SU24 remains relevant in S/4HANA because backend transactions and services still rely on authorization proposals.

For Fiori:

- **OData services**
- **Backend transactions**
- **Authorization objects**

should all be reviewed and maintained appropriately.

Custom Fiori applications should also have accurate SU24 proposals to simplify future role maintenance.

Q144. Fiori Performance Issue

Interviewer

"A Fiori app takes 40 seconds to load. Security team is blamed. What would you investigate?"

Executive Answer

I would avoid assuming Security is the root cause.

My investigation would include:

- 1. Browser developer tools.**
- 2. OData response time.**
- 3. Backend work process performance.**
- 4. CDS execution plans.**
- 5. Authorization trace (to identify repeated failed checks).**

6. HANA performance.
7. Network latency.
8. Fiori Launchpad performance.
9. ICF service health.
10. Basis and application logs.

Security should be evaluated as one possible factor, not the default assumption.

Q145. Real Project Scenario

Interviewer

"After transporting new Fiori roles to Production, users report that they can see the new tile but receive 'No Data Available.' What would you do?"

Executive Answer

I would troubleshoot systematically:

1. Confirm the tile launches successfully, indicating Launchpad content is assigned correctly.
2. Verify the OData service is active and functioning.
3. Run SU53 or STAUTHTRACE immediately after reproducing the issue.
4. Check backend authorization objects and organizational values such as Company Code, Plant, Purchasing Organization, or Sales Organization.
5. Validate CDS authorization (DCL) if the application retrieves data through CDS Views.
6. Confirm the underlying business data actually exists and meets the application's selection criteria.
7. Compare the affected user's authorizations with a user who can successfully view the data.
8. If the issue is authorization-related, update the role through the standard transport process rather than making direct production changes.

This layered approach isolates whether the issue lies in the UI, service layer, authorization model, or business data.

Advanced Interview Scenario

Interviewer

"How would you secure a global Fiori deployment across 25 countries?"

Executive Answer

I would design the solution using a standardized global architecture with localized organizational restrictions.

1. Define global business personas.
2. Create reusable business catalogs.
3. Use master and derived backend roles to manage country-specific organizational values.
4. Implement Spaces and Pages based on business functions rather than countries.
5. Secure OData services and ICF nodes.
6. Apply CDS authorization with DCL to enforce row-level access.
7. Validate the design through SAP GRC ARA to identify SoD conflicts.
8. Integrate with IAS/IPS or Microsoft Entra ID for centralized authentication and provisioning.
9. Establish governance processes for periodic role reviews, catalog maintenance, and audit reporting.

This approach balances consistency, scalability, and local compliance requirements.

Architect's Insight

One of the biggest misconceptions is:

"Fiori Security is just Catalogs and OData."

That's incorrect.

A complete Fiori security model spans:

Authentication



Launchpad



Catalogs



Spaces



Pages



Target Mapping



OData Services



ICF Nodes



Backend Roles



Authorization Objects



CDS Views



DCL



Business Data

A failure at any layer can prevent the application from functioning or expose data incorrectly.

End of Chapter 3

You now have a comprehensive understanding of:

- **CDS Views**
- **CDS Authorization**
- **DCL**
- **Embedded Analytics**
- **RAP Authorization**
- **Custom Fiori Security**

- **Launchpad Content Manager**
 - **Advanced Role Design**
 - **Fiori Performance Troubleshooting**
 - **Architect-level implementation strategies**
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Next Major Category – SAP Identity Management (SAP IDM), IAS, IPS & Identity Governance

Given your background in SAP IDM and the "Preferred Skills" in your target job description, this section will be one of your strongest differentiators. We'll cover:

- **SAP IDM Architecture**
- **Identity Center**
- **Runtime Engine**
- **Repository**
- **Passes & Jobs**
- **Provisioning Framework**
- **Active Directory & LDAP Integration**
- **IAS & IPS**
- **Joiner/Mover/Leaver**
- **Identity Governance**
- **Troubleshooting**
- **Migration from SAP IDM to SAP Cloud Identity/IAG**
- **Real project scenarios and architect-level interview questions**

This category is particularly valuable because relatively few candidates have deep SAP IDM experience, making it an opportunity to stand out in senior interviews.